

Error analysis motivates a staged pipeline that reduces unsupported ATT&CK predictions

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Motivation

Cyber threat intelligence (CTI) reports describe how attackers behave during cyber intrusions. Defenders read them to recognize and block similar intrusions. To compare threats across reports, analysts hand-label each described behavior with a MITRE ATT&CK technique ID — a standardized identifier for a specific adversary behavior, such as capturing screenshots. This labeling is slow expert work. Language models could automate it, but they often name techniques the report never describes and miss ones it does.

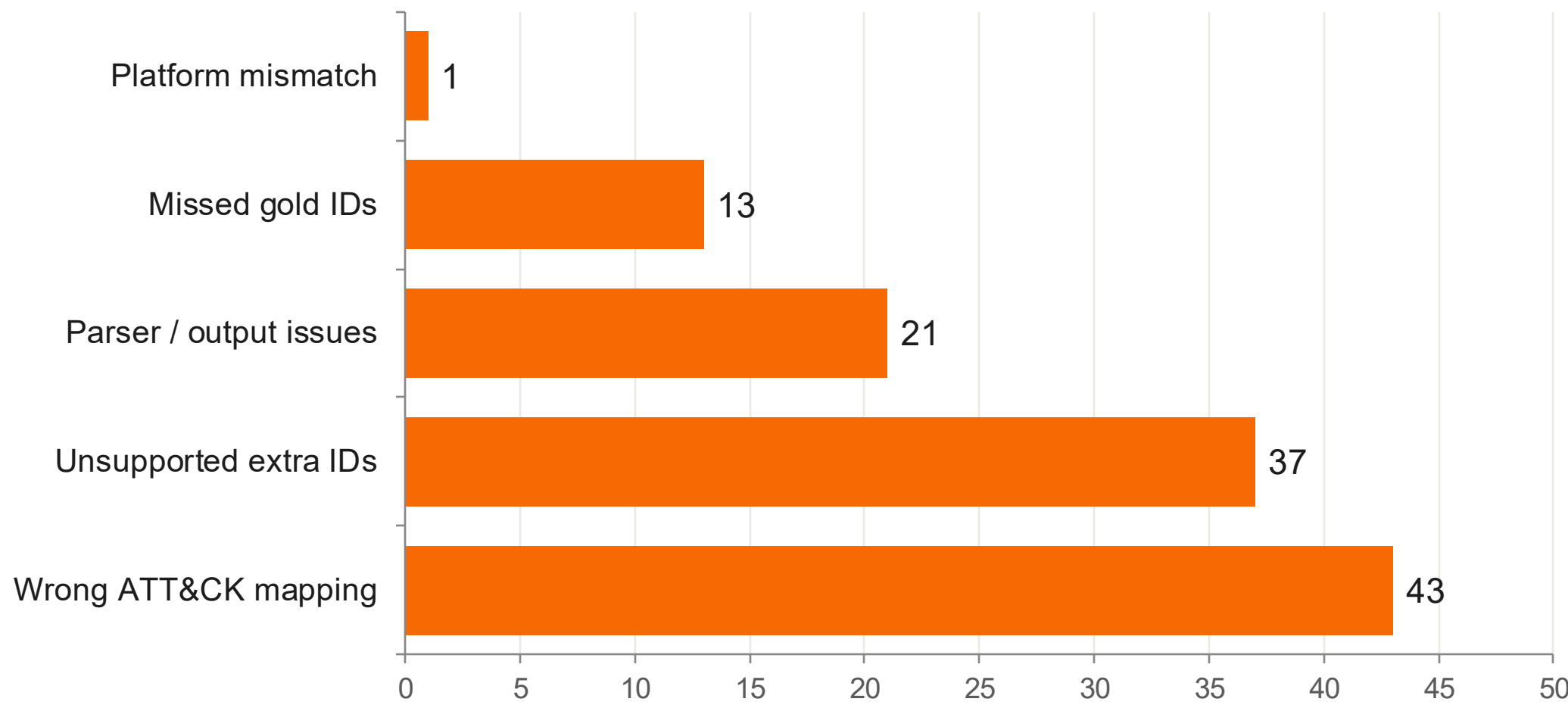
What kinds of LLM errors most strongly influence micro-F1 in ATT&CK technique extraction, and does targeting an observed failure mode measurably improve extraction?

Methods

Benchmark CTIBench CTI-ATE — 60 threat reports (47 Enterprise, 13 Mobile) with 397 expert-assigned technique IDs
Model Llama 3.1 8B Instruct, run locally at temperature 0 (no randomness)
Reproducibility The baseline was run five times; all five gave identical scores (SD = 0.0000)
Scoring Micro-precision, micro-recall and micro-F1 pooled over all 60 reports; per-example F1 also recorded
Parser diagnostic A check of whether the scoring code was affecting the results. Although it changed the scores slightly, unsupported ATT&CK technique predictions remained the dominant issue, motivating the staged pipeline.

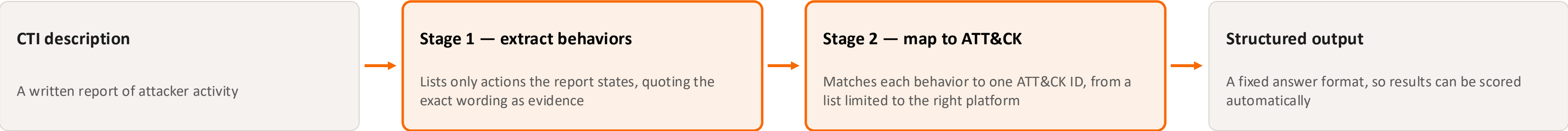
Observed Failure Modes

All 60 reports were hand-coded by error type. Wrong ATT&CK mapping means the model spotted the right behavior but named the wrong technique. Unsupported extra IDs are techniques it predicted with nothing in the report to support them. These two dominated, and motivated the staged design.

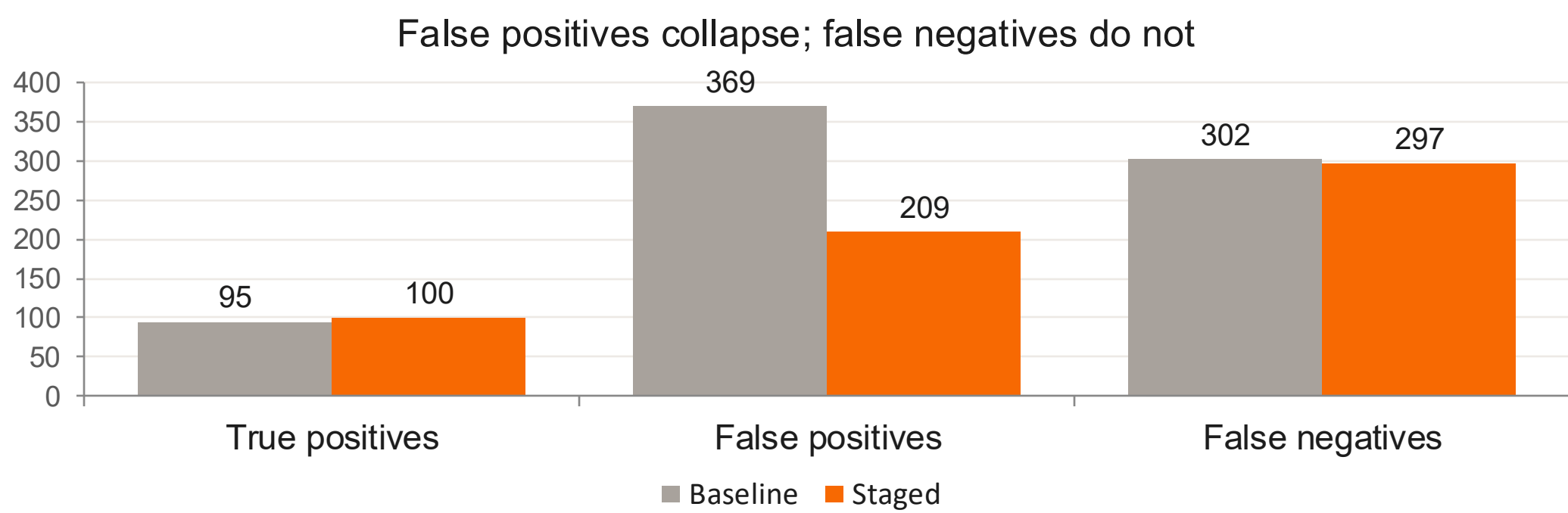
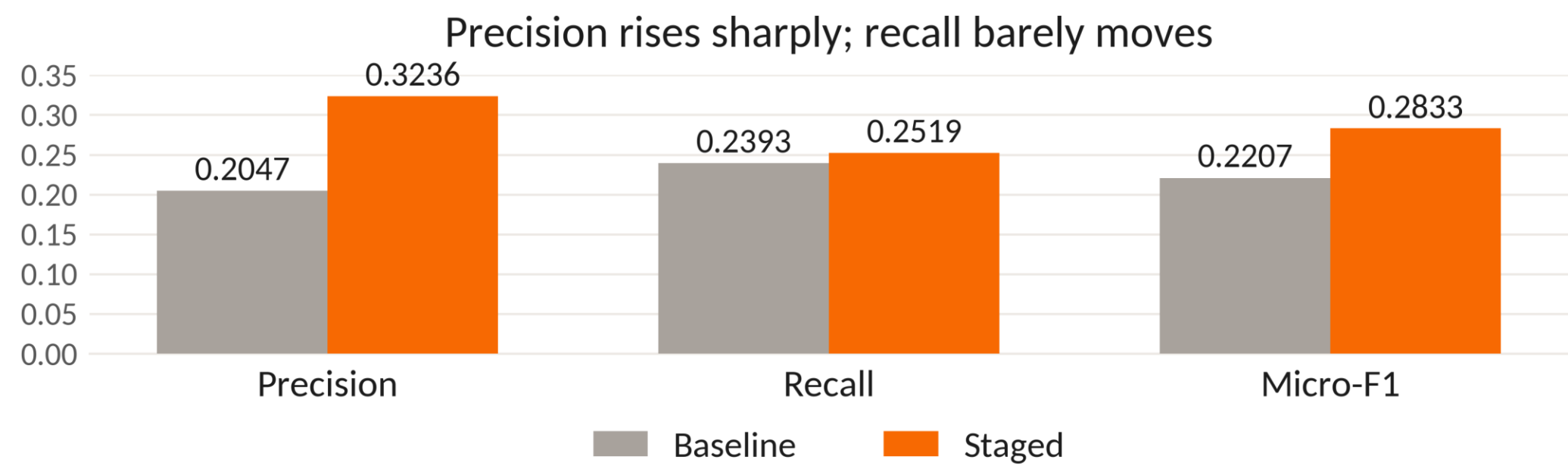


Categories overlap; some examples carry a primary and a secondary label.

Evidence-Grounded Staged Extraction



Method	Precision	Recall	Micro-F1	TP	FP	FN
Baseline	0.2047	0.2393	0.2207	95	369	302
Evidence-grounded staged	0.3236	0.2519	0.2833	100	209	297



Precision increased by 58%, while recall rose only slightly, from 0.2393 to 0.2519. The project’s evidence-grounded design required quoted support before assigning an ATT&CK technique, reducing unsupported technique predictions by 43% (369 → 209). The improvement came mainly from fewer false positives rather than recovering missed techniques; false negatives decreased only from 302 to 297.

Paired Error Analysis

19 of 60 examples, manually coded — 18 coded high-confidence

16 of 19

residual errors traced to Stage 2 mapping

- Both stages returned correctly formatted output on all 60 reports
- In the baseline the model mostly invented techniques and mislabeled the ones it did find
- After staging, remaining errors were confusion between similar techniques and missed behaviors
- Behaviors were often identified correctly, then assigned the wrong technique ID
- Formatting problems persisted in 2 of the 19 reports

Limitations

- One model, temperature 0, and 60 examples.
- Improvements were not uniform across examples.
- Average per-example F1 increased only from 0.2460 to 0.2651, less than the relative micro-F1 gain.
- Absolute performance remained low.
- Paired error analysis covered 19 of 60 examples.

Conclusions

Requiring the model to quote evidence before naming a technique cut unsupported ATT&CK predictions by 43% (369 → 209) and raised micro-F1 from 0.2207 to 0.2833. The gain came almost entirely from precision (0.2047 → 0.3236): the model became more selective, not more thorough. Recall barely moved (0.2393 → 0.2519).

Within the 19-example paired analysis, Stage 1 usually found the right behaviors while Stage 2 assigned the wrong technique ID. Evidence grounding fixes overprediction; it does not fix mapping. These are separate problems, and Stage 2 mapping is the clearest opportunity for future work.

Future Work

- Target Stage 2 mapping directly — the dominant residual error
- Address false negatives, now the largest remaining error source
- Evaluate mapping-focused methods on the full 60 examples
- Report micro-F1 alongside per-example F1

EXPLORATORY — NOT A VALIDATED RESULT

Retrieval diagnostics: when the system offers the model k candidate techniques to choose from, the correct one is usually among them (0.4253 at k=8; 0.6667 at k=25; 0.8046 at k=40). A bounded 3-example test showed no improvement, and the full 60-report run is paused pending validation. Having the right candidate available does not mean the model picks it.